# DeepSeek-V2: Efficient MoE Language Model with Multi-Head Latent Attention

**236B total params • 21B activated/token • 128K context**

Audience: ML researchers, NLP engineers, graduate students

Source: DeepSeek-V2 paper (arXiv:2405.04434)

# The LLM Scaling Dilemma: Performance vs. Training Cost and Inference Speed

## Bigger dense models improve capability—but costs scale aggressively.

Training bottlenecks
- Massive compute budget for dense scaling
- High cost to reach top-tier quality

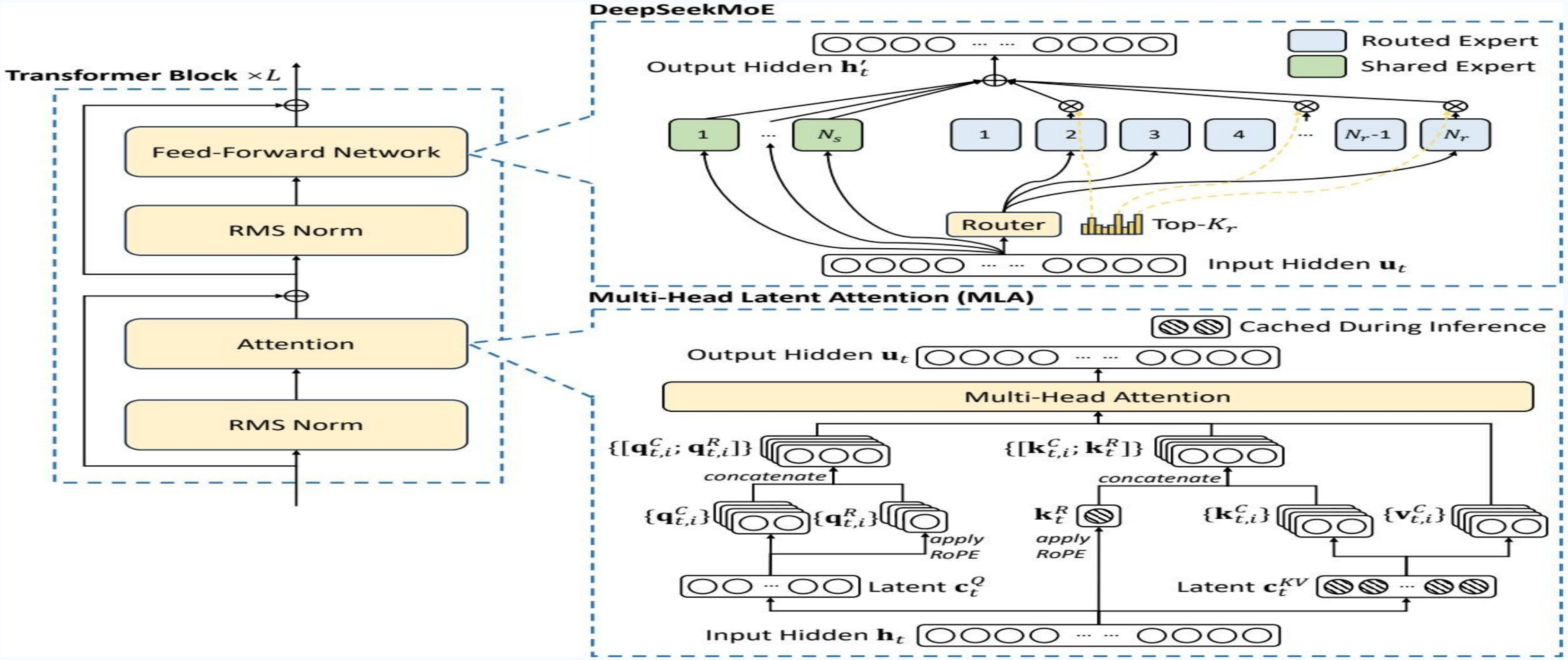
Inference bottlenecks
- KV cache grows with context length
- Autoregressive generation throughput drops
- Memory pressure limits deployment efficiency

DeepSeek-V2 strategy
1) MLA: compress attention KV states into latent form
2) DeepSeekMoE: activate only a sparse expert subset

Goal: shift the Pareto frontier of quality vs. efficiency.

# DeepSeek-V2 Architecture: MLA + DeepSeekMoE in a Transformer Block

## MLA replaces standard attention; DeepSeekMoE replaces dense FFN.

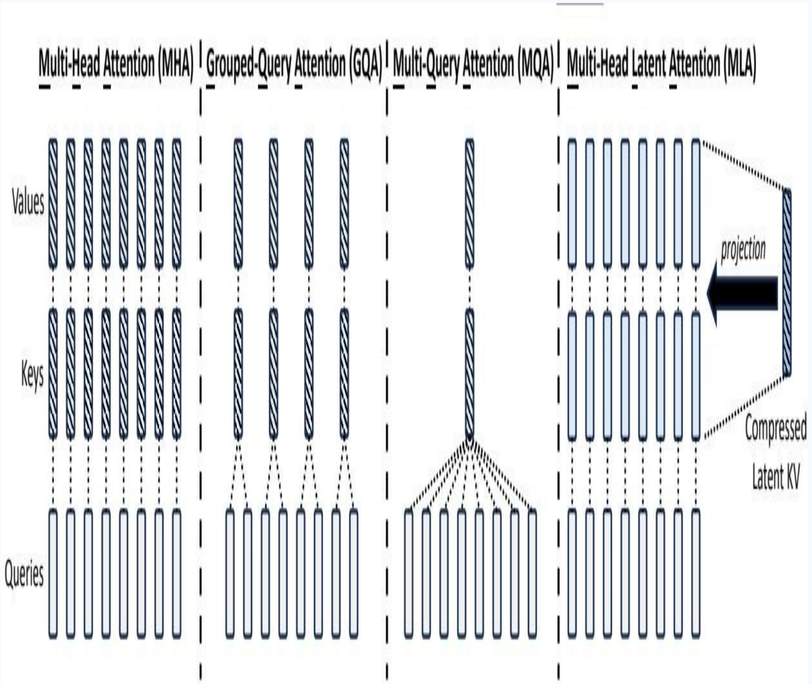



Model scale: 236B total parameters, 21B activated per token, 128K context.

# Multi-Head Latent Attention Compresses KV Cache by 93.3%

Core idea: low-rank KV joint compression

• Cache compact latent vector c_t^(KV), not full per-head K/V
• Reconstruct K, V and decoupled RoPE key on-the-fly
• Preserves or improves quality vs. standard MHA
• Strictly improves inference memory efficiency

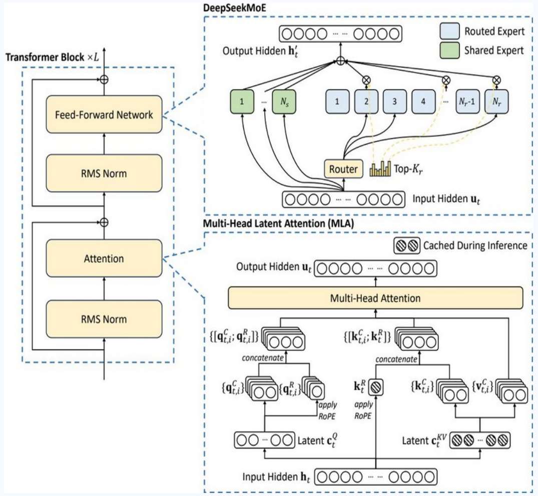



Attention alternatives: MHA, GQA, MQA, MLA (paper Fig. 2)

# DeepSeekMoE: Fine-Grained Experts with Shared Expert Isolation

**FFN uses two expert types: shared experts (always on) + routed experts (top-K selected).**

- Shared experts capture common knowledge across tokens
- Routed experts specialize via sparse token-dependent activation
- Fine-grained segmentation improves specialization capacity
- Device-limited routing controls cross-node communication
- Auxiliary losses balance load across experts/devices
- Token dropping during training prevents expert overflow (no inference impact)



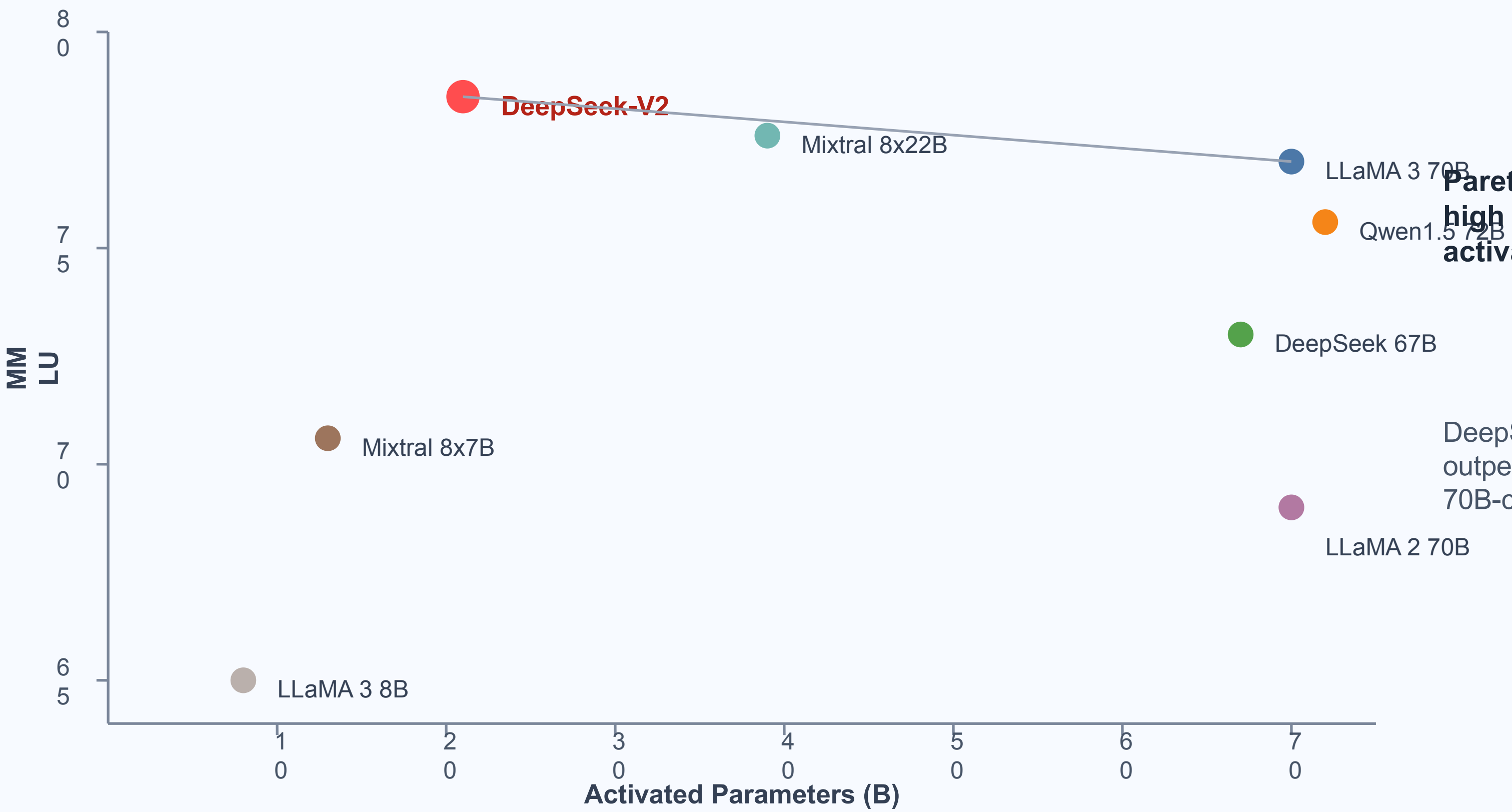
# 42.5% Training Cost Reduction and 5.76× Generation Throughput vs. DeepSeek 67B

**Relative efficiency comparison (DeepSeek 67B baseline → DeepSeek-V2):**

| Training Cost | KV Cache Size | Max Generation Throughput |
| --- | --- | --- |
| DeepSeek-V2 | DeepSeek-V2 | DeepSeek-V2 |
| **42.5% reduction** | **93.3% reduction** | **5.76× higher** |

*No absolute FLOPs/cache/throughput values shown; only source-reported relative gains.*

# DeepSeek-V2 Tops MMLU Among Open-Source Models at Only 21B Activated Parameters




**Pareto-optimal region: high MMLU at low activated parameters**

DeepSeek-V2 (21B, 78.5) outperforms or matches 70B-class dense models.

# DeepSeek-V2 Chat (RL) Rivals Closed-Source Models on Chinese Benchmarks

**Post-training: SFT on 1.5M conversational sessions + GRPO reinforcement learning**

| Benchmark | Score |
|---|---|
| AlpacaEval 2.0 (length-controlled win rate) | **38.9%** |
| MT-Bench (overall) | **8.97** |
| AlignBench (overall, Chinese) | **7.91** |

Claim from source brief: strongest open-source chat model in Chinese; competitive with closed-source systems in English open-ended evaluation.

# Key Takeaways: Architectural Innovation Unlocks Efficient Scaling

1. MLA is a stronger attention primitive for long-context inference efficiency (93.3% KV cache reduction).
2. DeepSeekMoE enables economical training of a 236B-parameter model (42.5% lower training cost vs DeepSeek 67B).
3. Only 21B activated parameters achieve top-tier open-source MMLU (78.5), matching/exceeding 70B-class dense peers.
4. Generation efficiency scales materially in deployment (5.76× max throughput vs DeepSeek 67B).
5. SFT + GRPO alignment yields strong multilingual chat quality, especially in Chinese benchmarks.

**DeepSeek-V2 sits on the quality-efficiency Pareto frontier for open-source LLMs.**